

ASSIGNMENT

Git, GitHub, GitLab and Visual Studio Code

Subject: Version Control and Development Tools

1. Introduction

Git is a distributed version-control system used to record and manage changes in source code. GitHub and GitLab are online platforms that host Git repositories and provide collaboration features. Visual Studio Code (VS Code) is a source-code editor with built-in Git support. These tools are commonly used together in modern software development.

2. Questions and Answers

Q1. What is Git?

Answer: Git is a distributed version-control system. It helps developers track changes in files, maintain different versions of a project, create branches, and collaborate safely.

Q2. Why is Git used?

Answer: Git is used to keep a history of project changes, restore earlier versions, work on features independently, and collaborate with other developers.

Q3. What is a Git repository?

Answer: A Git repository is a project directory that contains the files and the Git information required to track their history and changes.

Q4. What is GitHub?

Answer: GitHub is an online platform for hosting Git repositories. It supports collaboration, code review, pull requests, issues, project management, and other development features.

Q5. What is GitLab?

Answer: GitLab is a Git-based development platform that provides repository hosting, collaboration, issue tracking, and DevOps features such as continuous integration and continuous delivery.

Q6. What is VS Code?

Answer: Visual Studio Code is a lightweight source-code editor developed by Microsoft. It supports many programming languages and provides an integrated terminal and Source Control features.

Q7. What is the difference between Git and GitHub?

Answer: Git is the version-control software that runs on a computer. GitHub is an online service used to host Git repositories and collaborate with others.

Q8. What is the difference between GitHub and GitLab?

Answer: Both are platforms for hosting Git repositories and collaborating on software projects. GitLab also has a strong focus on integrated DevOps and CI/CD workflows.

Q9. What is a commit?

Answer: A commit is a saved snapshot of changes in a Git repository. A commit normally includes a message describing what was changed.

Q10. What is a branch?

Answer: A branch is an independent line of development. Developers can create branches to work on features or fixes without immediately changing the main branch.

Q11. What is staging in Git?

Answer: Staging means selecting changes that will be included in the next commit. The command `git add` is commonly used to stage changes.

Q12. What is a remote repository?

Answer: A remote repository is a Git repository stored somewhere outside the local computer, such as on GitHub or GitLab.

Q13. What does `git clone` do?

Answer: The `git clone` command creates a local copy of an existing remote repository and downloads its files and Git history.

Q14. What does `git push` do?

Answer: The `git push` command sends local commits to a remote repository such as GitHub or GitLab.

Q15. What does `git pull` do?

Answer: The `git pull` command downloads changes from a remote repository and integrates them into the current local branch.

3. Important Git Commands

```
git --version – Checks the installed Git version.
git config --global user.name "Your Name" – Sets the Git username.
git config --global user.email "your@email.com" – Sets the Git email address.
git init – Creates a new Git repository.
git status – Shows the current state of the working directory.
git add . – Stages all changed and untracked files.
git commit -m "message" – Creates a commit with a descriptive message.
git log --oneline – Displays a short version of the commit history.
git branch – Lists local branches.
git switch -c feature – Creates and switches to a new branch.
git switch main – Switches to the main branch.
git merge feature – Merges the feature branch into the current branch.
git remote -v – Displays configured remote repository URLs.
git push – Uploads local commits to a remote repository.
git pull – Downloads and integrates remote changes.
git clone – Creates a local copy of a remote repository.
```

4. Practical Example: Git + GitHub

A basic workflow for creating a project locally and uploading it to GitHub is:

```
mkdir student-project
cd student-project
git init
echo "# Student Project" > README.md
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin https://github.com/USERNAME/student-project.git
git push -u origin main
```

5. Practical Example: Branching

```
git switch -c login-feature
# Make changes to the project
git add .
git commit -m "Added login feature"
git switch main
git merge login-feature
git push origin main
```

6. GitHub and GitLab Questions

Q16. How do you connect a local Git repository to GitHub?

Answer: Use git remote add origin followed by the URL of the GitHub repository, then push the local branch using git push.

Q17. How do you clone a GitLab repository?

Answer: Use git clone followed by the GitLab repository URL.

Q18. Why are remote repositories useful?

Answer: They provide online storage for project history and make collaboration, backup, code review, and sharing easier.

Q19. What is a pull request?

Answer: A pull request is a request to merge changes from one branch into another. It allows changes to be reviewed before they are merged.

Q20. What is CI/CD in GitLab?

Answer: CI/CD stands for Continuous Integration and Continuous Delivery or Deployment. It automates tasks such as building, testing, and deploying software.

7. VS Code Questions and Answers

Q21. How can Git be used inside VS Code?

Answer: VS Code provides a Source Control panel and integrated terminal through which users can stage files, commit changes, create branches, push changes, and pull changes.

Q22. How do you open a project in VS Code from the terminal?

Answer: After navigating to the project folder, use the command code .

Q23. What is the Source Control panel?

Answer: It is the VS Code interface for viewing changes and performing Git operations such as staging, committing, branching, and synchronization.

Q24. What is the advantage of using VS Code with Git?

Answer: It allows developers to edit code and manage version control from the same application, making the development workflow faster and easier.

8. Short-Answer Questions

1. Define version control.
2. What is the purpose of git init?
3. What is the purpose of git add?
4. What is the purpose of git commit?
5. What is the difference between a local and remote repository?
6. What is a Git branch?
7. What is GitHub used for?
8. What is GitLab used for?
9. What is VS Code?
10. What is the purpose of git pull?

9. Practice Questions

The following practical questions are intended for students to perform on their own computer. Write the commands you use and record the output where appropriate.

1. Install Git and find the installed Git version.
2. Configure your Git username and email.
3. Create a new folder named college-project and initialize it as a Git repository.
4. Create a README.md file and add a short description of your project.
5. Check the repository status before and after staging the file.
6. Stage the README.md file and create a commit named "Added README".
7. Create a new branch named feature-1 and switch to it.
8. Create a second file on feature-1, commit it, and display the commit history.
9. Create a repository on GitHub and connect your local project to it.
10. Push the main branch to GitHub.
11. Create another branch, make a change, commit it, and push the branch.
12. Merge the branch into main and push the updated main branch.
13. Clone a GitHub or GitLab repository into a new folder.
14. Open the cloned project in VS Code and use the Source Control panel.
15. Explain the difference between git push and git pull with an example.
16. Create a GitLab project and upload a local Git repository to it.

10. Viva / Oral Questions

1. What is Git?

2. Why do developers use version control?
3. What is the difference between Git and GitHub?
4. What is the difference between GitHub and GitLab?
5. What is a commit?
6. What is a branch?
7. What is a merge?
8. What is a remote repository?
9. What does git clone do?
10. What does git push do?
11. What does git pull do?
12. How does VS Code help with Git?

11. Conclusion

Git, GitHub, GitLab, and VS Code are important tools in modern software development. Git provides version control, while GitHub and GitLab provide online collaboration and repository hosting. VS Code provides a convenient environment for writing code and performing Git operations. Learning these tools helps students understand professional software-development workflows and collaborative programming.